STAPLES EDUCATION · ODE QUICK REFERENCE

First-Order ODE Handout

Formula Sheet, Decision Tree, Method Summaries

COLOR-CODING KEY

Color	Meaning	Where Used
Black	Explanatory prose, definitions, quoted problems	All body text
Blue (#1f61c9)	Structural headers, eyebrows, step labels	Section titles, STEP labels
Purple (#6B2FA0)	Live mathematics being derived step-by-step	All derivation equations
Orange (#E07B00)	Final answers in boxed callouts	Boxed FINAL ANSWER

FORMULA SHEET

GENERAL FORM

$$dy/dx = f(x, y) \quad \text{or} \quad M\,dx + N\,dy = 0$$

1. SEPARATION OF VARIABLES

When: $f(x,y) = g(x)\cdot h(y)$ (factors into pure-x times pure-y)

$$dy/h(y) = g(x)\,dx \rightarrow \int (1/h(y))\,dy = \int g(x)\,dx + C$$

Check: $h(y) = 0$ gives equilibrium solutions.

2. LINEAR FIRST-ORDER ODE

When: Can be written as $y' + P(x)y = Q(x)$

$$\mu(x) = e^{\int P(x)\,dx} \text{ (drop the +C)}$$

$$\text{Multiply ODE by } \mu\text{: } d/dx[\mu\cdot y] = \mu\cdot Q \rightarrow \mu\cdot y = \int \mu\cdot Q\,dx + C$$

3. EXACT EQUATIONS

When: $M\,dx + N\,dy = 0$ and $\partial M/\partial y = \partial N/\partial x$

$$F = \int M\,dx + g(y)\text{, then match } F_y = N \text{ to find } g'(y)\text{, integrate to get } g(y)$$

$$\text{Solution: } F(x, y) = C$$

4. NON-EXACT + INTEGRATING FACTOR

When: $M_y \neq N_x$, but one of the tests below yields a single-variable function

$$\text{Test 1 (x-only): } (M_y - N_x)/N = f(x) \rightarrow \mu(x) = e^{\int f(x)\,dx}$$

$$\text{Test 2 (y-only): } (N_x - M_y)/M = g(y) \rightarrow \mu(y) = e^{\int g(y)\,dy}$$

Multiply original by μ, verify new equation is exact, then solve as Exact.

5. NUMERICAL METHODS

When: ODE is unsolvable, infeasible, or inefficient to solve analytically

$$\text{Euler } (O(h) \text{ global}): y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

$$\text{Heun } (O(h^2) \text{ global}): \tilde{y} = y_n + h \cdot f(x_n, y_n);\ y_{n+1} = y_n + (h/2)[f(x_n, y_n) + f(x_{n+1}, \tilde{y})]$$

$$\text{RK4 } (O(h^4) \text{ global}): k_1 = f(x_n, y_n);\ k_2 = f(x+h/2, y+h \cdot k_1/2);\ k_3 = f(x+h/2, y+h \cdot k_2/2);\ k_4 = f(x+h, y+h \cdot k_3)$$

$$y_{n+1} = y_n + (h/6)(k_1 + 2k_2 + 2k_3 + k_4)$$

DECISION TREE

Given dy/dx = f(x,y) or M dx + N dy = 0, ask these questions in strict hierarchical order. Do not guess — work through the tree top to bottom.

Step	Question	If YES...	If NO...
1	Does f(x,y) factor into g(x)·h(y)?	**SEPARABLE. Separate and integrate. Check h(y)=0 for equilibria.**	Go to Step 2.
2	Can it be written as y' + P(x)y = Q(x)?	**LINEAR. Compute μ = e^∫P dx, multiply, collapse product, integrate.**	Go to Step 3.
3	In form M dx + N dy = 0, is M_y = N_x?	**EXACT. Find potential F(x,y) = C by integrating M in x, matching F_y = N.**	Go to Step 4.
4	Is (M_y - N_x)/N a function of x alone?	**IF μ(x) = e^∫f(x) dx. Multiply, verify exact, solve as Exact.**	Try Step 5.
5	Is (N_x - M_y)/M a function of y alone?	**IF μ(y) = e^∫g(y) dy. Multiply, verify exact, solve as Exact.**	Go to Step 6.
6	Need a numerical value but no closed form?	**NUMERICAL. Use RK4 (default), Heun (moderate), or Euler (quick).**	Try substitution (Bernoulli, homogeneous) or consult a re

Note: Many equations fit multiple categories. A homogeneous linear ODE is also separable. A separable equation written in differential form is also exact. Always choose the path of least cognitive resistance — the method that requires the fewest algebraic manipulations.

METHOD 1 — SEPARATION OF VARIABLES

Formula: $dy/dx = g(x)\cdot h(y) \rightarrow \int(1/h(y))\,dy = \int g(x)\,dx + C$

When to use: The ODE factors into a pure function of x times a pure function of y.

Worked example (3 steps):

```
Solve: dy/dx = 2xy, y(0) = 1
Step 1 — Separate: dy/y = 2x dx
Step 2 — Integrate: ln|y| = x² + C
Step 3 — Apply IC: 1 = e⁰ → C=0
y(x) = e^(x²)
```

METHOD 2 — LINEAR FIRST-ORDER ODE

Formula: $y' + P(x)y = Q(x) \to \mu = e^{\int P\,dx} \to d/dx[\mu y] = \mu Q$

When to use: y and y' appear linearly (first power, not multiplied together).

Worked example (3 steps):

Solve: $y' + (2/x)y = 4x$, $y(1) = 2$

Step 1 — $\mu = e^{\int (2/x)\,dx} = x^2$

Step 2 — Multiply: $d/dx[x^2 y] = 4x^3 \to x^2 y = x^4 + C$

Step 3 — Apply IC: $2 = 1 + C \to C = 1$

$y(x) = x^2 + 1/x^2$

METHOD 3 — EXACT EQUATIONS

Formula: $M\,dx + N\,dy = 0$, $M_y = N_x \rightarrow F = \int M\,dx + g(y)$, match $F_y = N$

When to use: The ODE is in differential form and the mixed partials are equal.

Worked example (3 steps):

$$\text{Solve: } (2xy + 3)\,dx + (x^2 - 1)\,dy = 0$$

$$\text{Step 1 — Test: } M_y = 2x = N_x \ ✓ \text{ Exact}$$

$$\text{Step 2 — } F = \int(2xy+3)\,dx = x^2y + 3x + g(y)$$

$$\text{Step 3 — } F_y = x^2 + g'(y) = x^2 - 1 \rightarrow g(y) = -y$$

$$\mathbf{x^2y + 3x - y = C}$$

METHOD 4 — NON-EXACT + INTEGRATING FACTOR

Formula: If $M_y \neq N_x$, test $(M_y - N_x)/N$ for x-only or $(N_x - M_y)/M$ for y-only

When to use: The ODE is in differential form but not exact — needs an IF to force exactness.

Worked example (3 steps):

```
Solve: (2x² - y) dx + (x²y + x) dy = 0
Step 1 — (M_y - N_x)/N = -2/x → μ(x) = e^∫(-2/x) dx = 1/x²
Step 2 — Multiply: (2 - y/x²)dx + (y + 1/x)dy = 0. Verify exact ✓
Step 3 — F = 2x + y/x + y²/2
2x + y/x + y²/2 = C
```

METHOD 5 — NUMERICAL METHODS

Formula: $y_{n+1} = y_n + (h/6)(k_1 + 2k_2 + 2k_3 + k_4)$ [RK4]

When to use: ODE has no closed-form solution, or the solution is impractical to evaluate.

Worked example (RK4, one step):

```
Solve: dy/dx = y, y(0) = 1, h = 0.1 (one step)
k₁ = f(0, 1) = 1.0000
k₂ = f(0.05, 1.05) = 1.0500
k₃ = f(0.05, 1.0525) = 1.0525
k₄ = f(0.1, 1.10525) = 1.1053
y₁ = 1 + (0.1/6)(1 + 2.1 + 2.105 + 1.1053) ≈ 1.10517
y(0.1) ≈ 1.10517 (Exact: e^0.1 = 1.10517, error < 10⁻⁷)
```

ERROR COMPARISON

Method	Slope Evals/Step	Local Error	Global Error	Best For
Euler	1	$O(h^2)$	$O(h)$	Quick estimates, teaching
Heun	2	$O(h^3)$	$O(h^2)$	Moderate accuracy, smooth ODEs
RK4	4	$O(h^5)$	$O(h^4)$	General-purpose, industry default